CHAPTER – 1

INTRODUCTION

1.1 Overview of the Project

The proposed project, titled **"**Weather Balloon using IoT-Based Monitoring System**,"** focuses on the design and development of a compact, low-cost, and wireless atmospheric data acquisition system. The system is intended to be carried by a weather balloon to measure key environmental parameters at varying altitudes and transmit them in real time to a remote monitoring platform.

At the core of the system lies the ESP32-WROOM-32, a powerful and energy-efficient microcontroller equipped with built-in Wi-Fi capabilities. This module acts as the central processing unit of the payload, responsible for interfacing with sensors, processing the collected data, and transmitting it to cloud servers. Its integrated wireless communication eliminates the need for additional communication hardware, thereby reducing system complexity and overall weight--an important factor in balloon-based applications.

The payload is equipped with multiple sensors to capture a comprehensive set of atmospheric parameters:

- The BMP280 sensor is used to measure temperature, atmospheric pressure, and altitude. By utilizing pressure variations, the system can estimate altitude changes as the balloon ascends.
- The DHT11 sensor is used to measure relative humidity, providing insights into moisture content in the air.
- The MQ-135 gas sensor is employed to monitor air quality, including parameters such as Air Quality Index (AQI), carbon dioxide (CO_2), and ammonia (NH_3) levels.

Together, these sensors enable multi-parameter environmental monitoring, making the system suitable for basic atmospheric studies and experimental analysis.

The entire system is powered by a 3.7V Li-ion rechargeable battery, which ensures portability and lightweight operation. However, since the ESP32 and sensors require a stable higher voltage for optimal performance, the system incorporates DC-DC boost converters (XL6009 and MT3608) to step up the voltage to required levels. Additionally, a TP4056 charging module is integrated to allow safe and efficient recharging of the battery. Supporting components such as 10kΩ resistors and a 470μF capacitor are used for signal conditioning and voltage stabilization, ensuring reliable circuit operation under varying conditions.

Once the system is powered on, the ESP32 initializes all connected sensors and begins collecting environmental data at predefined intervals. The acquired data is then processed and transmitted via Wi-Fi to the **Blynk cloud server. This cloud-based approach enables real-time data storage, remote accessibility, and visualization through a user-friendly mobile application. Users can monitor atmospheric conditions live, without needing to physically retrieve the payload.

The working concept of the system can be summarized as follows:

This architecture demonstrates the integration of embedded systems, sensor technology, power electronics, and IoT communication into a single cohesive platform. The project emphasizes simplicity, affordability, and accessibility, making it particularly suitable for academic purposes, student projects, and preliminary environmental research.

1.2 Motivation of Project

The motivation behind this project stems from the need for a low-cost and accessible atmospheric monitoring system. Traditional weather monitoring methods, such as radiosondes and satellite systems, are highly accurate but involve significant cost, complex infrastructure, and limited accessibility for students and small research groups. This creates a gap between theoretical knowledge and practical implementation in environmental and embedded systems engineering.

With the advancement of IoT technologies, it has become possible to design compact and efficient systems using microcontrollers like the ESP32-WROOM-32. These systems enable real-time data acquisition and wireless transmission without requiring expensive equipment. The integration of cloud platforms such as Blynk further enhances usability by allowing monitoring through mobile devices.

Another key motivation is the increasing importance of environmental monitoring. Parameters such as temperature, humidity, pressure, and air quality are essential for understanding climate conditions and pollution levels. This project provides a practical approach to collect such data in real time.

Additionally, the project offers hands-on experience in embedded systems, sensor integration, and IoT communication. Overall, it aims to demonstrate that effective atmospheric monitoring can be achieved using affordable, portable, and scalable technology.

1.3 Problem Statement

Accurate atmospheric data collection is essential for understanding weather patterns, climate change, and environmental conditions. However, conventional systems such as radiosondes and satellite-based monitoring solutions are expensive, complex, and require specialized infrastructure, making them inaccessible for academic institutions and small-scale research applications. Additionally, many existing systems do not provide easy real-time access to data for end users, limiting their practical usability.

There is a need for a compact, cost-effective, and user-friendly solution that can measure multiple atmospheric parameters and transmit data wirelessly for remote monitoring. The challenge lies in integrating sensors, power management, and communication modules into a lightweight system suitable for balloon deployment, while maintaining reliability and efficiency.

This project addresses these issues by developing an IoT-based weather balloon system using the ESP32-WROOM-32 and cloud platforms like Blynk to enable real-time, accessible atmospheric monitoring.

1.4 Project Objectives

The primary objective of this project is to design and develop a low-cost, lightweight, and IoT-enabled weather balloon system capable of monitoring atmospheric conditions and transmitting data in real time. The system is built using the ESP32-WROOM-32 and integrated sensors to achieve reliable environmental data acquisition.

Specific Objectives

- To design a compact payload system suitable for weather balloon deployment
- To measure key atmospheric parameters, including:
    - Temperature
    - Humidity
    - Pressure
    - Altitude
    - Air Quality (AQI, CO_2, NH_3)
- To interface multiple sensors (BMP280, DHT11, MQ-135) with the ESP32 for data acquisition
- To implement wireless communication using Wi-Fi for real-time data transmission
- To send collected data to cloud servers using the Blynk platform
- To enable remote monitoring and visualization of environmental data through a mobile application
- To ensure system stability and reliability using proper circuit components such as resistors and capacitors
- To develop a scalable system that can be enhanced with additional modules like GPS or long-range communication
- To provide a practical implementation platform for learning embedded systems, IoT, and sensor integration.

1.5 Scope and Limitations

The scope of this project includes the design and implementation of a low-cost IoT-based weather balloon system capable of measuring atmospheric parameters such as temperature, humidity, pressure, altitude, and air quality. The system utilizes the ESP32-WROOM-32 for data processing and Wi-Fi communication, enabling real-time transmission of sensor data to cloud platforms like Blynk. It is primarily intended for educational, experimental, and small-scale environmental monitoring applications.
However, the system has certain limitations. The reliance on Wi-Fi restricts operational range and may not be suitable for very high altitudes. Sensor accuracy is limited, particularly with DHT11 and MQ-135, which require calibration. Power constraints due to battery capacity may limit operational time. Additionally, environmental factors such as extreme temperature and pressure conditions can affect system performance and reliability during flight.

1.6 Alignment with SDGs

This project aligns with several **United Nations Sustainable Development Goals (SDGs)** by promoting innovation, environmental awareness, and accessible technology:

- **SDG 9 – Industry, Innovation, and Infrastructure:** Encourages the development of innovative and cost-effective technologies using embedded systems like the ESP32-WROOM-32, supporting technological advancement and research.
- **SDG 11 – Sustainable Cities and Communities:** Enables monitoring of environmental parameters such as air quality and humidity, contributing to improved urban planning and healthier living conditions.
- **SDG 12 – Responsible Consumption and Production:** Promotes efficient use of electronic components and encourages sustainable engineering practices through low-power system design.

- **SDG 13 – Climate Action:** Supports the collection of real-time atmospheric data, which can aid in understanding climate patterns and environmental changes.
- **SDG 4 – Quality Education:** Provides hands-on learning in IoT, embedded systems, and environmental monitoring, enhancing technical education and practical knowledge.

1.7 Organization of Report

This report is structured to systematically present the design, development, and analysis of the IoT-based weather balloon system:

- **Chapter 1 - Introduction :** This chapter provides an overview of the project, including its background, motivation, problem statement, objectives, scope, limitations, and alignment with Sustainable Development Goals (SDGs).
- **Chapter 2 – Literature Survey:** This chapter reviews existing atmospheric monitoring systems and IoT-based solutions. It highlights their advantages, limitations, and the need for the proposed system.
- **Chapter 3 – Proposed System:** This chapter explains the overall system architecture, hardware components such as the ESP32-WROOM-32 and sensors, circuit design, and software implementation.
- **Chapter 4 – Results and Analysis:** This chapter describes the practical implementation of the project, working procedure, data collection, and results obtained through the Blynk application.
- **Chapter 5 – Conclusion and Future Scope:** This chapter summarizes the outcomes of the project and suggests possible improvements and future enhancements.

CHAPTER-2

LITERATURE SURVEY

2.1 Review of Existing Systems

Existing large-scale weather balloon systems, commonly known as radiosondes, are widely used by meteorological organizations to collect atmospheric data. These systems are typically attached to high-altitude balloons filled with helium or hydrogen and are capable of measuring parameters such as temperature, pressure, humidity, and wind speed as they ascend through the atmosphere. The collected data is transmitted to ground stations using radio communication, enabling accurate weather forecasting and climate analysis.

These systems are highly reliable and provide precise measurements even at very high altitudes, often reaching the stratosphere. They are regularly deployed by weather agencies around the world and play a critical role in modern meteorology. However, despite their accuracy and importance, these systems have several limitations.

The major drawback of large weather balloon systems is their high cost and complexity. They require specialized equipment, trained personnel, and dedicated ground receiving stations. Additionally, they are not easily accessible for educational use or small-scale research projects. Maintenance and operational costs are also significant.Overall, while existing weather balloon systems are highly effective and accurate, they are not suitable for low-cost or student-level applications. This creates a need for simpler, more affordable alternatives like IoT-based weather monitoring systems.

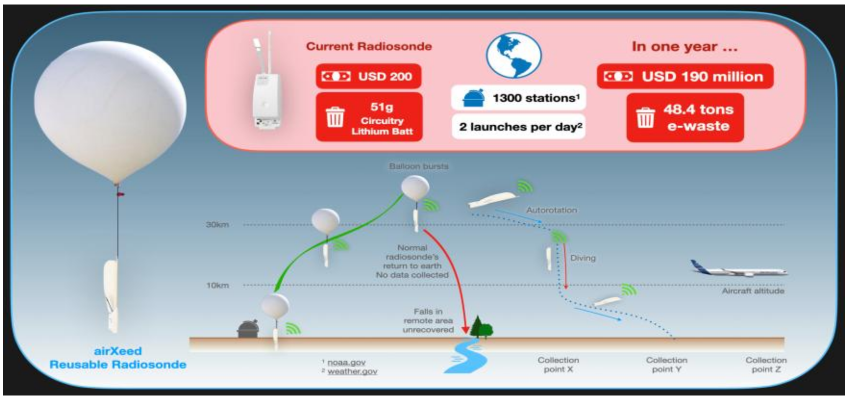

Fig- 2.1: Existing Weather balloon

2.2 Limitations of Existing Approaches

Existing atmospheric monitoring systems, while effective, present several limitations that restrict their widespread use, particularly in academic and small-scale applications. Traditional systems such as radiosondes provide accurate data but involve high costs, making them unsuitable for low-budget projects. Additionally, they require specialized receiving equipment and trained personnel, increasing operational complexity.

Satellite-based monitoring systems offer extensive coverage but are associated with very high infrastructure and maintenance costs. These systems are not directly accessible for hands-on experimentation and are typically limited to government or large research organizations.

Ground-based weather stations, although relatively affordable, are limited to fixed geographical locations and cannot provide vertical atmospheric data, which is essential for analysing altitude-based variations.

Recent IoT-based systems using microcontrollers like the ESP32-WROOM-32 improve affordability but still face challenges such as limited communication range, dependency on Wi-Fi connectivity, and potential data loss in remote areas. Platforms like Blynk require stable internet connectivity, which may not always be available during high-altitude deployments.

These limitations highlight the need for a more accessible, portable, and cost-effective solution.

2.3 Need for Proposed System

The limitations of existing atmospheric monitoring systems highlight the need for a cost-effective, portable, and user-friendly solution that can be easily implemented in academic and small-scale research environments. Traditional systems such as radiosondes and satellite-based monitoring are not only expensive but also require complex infrastructure and specialized expertise, making them impractical for students and low-budget projects.

There is a growing demand for systems that can provide real-time environmental data with minimal setup and operational complexity. In addition, the increasing importance of monitoring parameters such as temperature, humidity, pressure, altitude, and air quality for climate studies and pollution analysis further emphasizes the need for accessible solutions.

The proposed system addresses these challenges by utilizing modern IoT technology with the ESP32-WROOM-32, enabling efficient data processing and wireless communication. By integrating cloud platforms like Blynk, the system allows users to monitor atmospheric conditions remotely in real time through a mobile application.

Furthermore, the proposed design is lightweight, scalable, and economical, making it suitable for weather balloon deployment. It also provides a practical platform for learning embedded systems, sensor integration, and IoT communication, thereby fulfilling both technical and educational requirements.

CHAPTER-3
PROPOSED SYSTEM

3.1 Introduction

The proposed system is designed as an integrated embedded architecture for real-time atmospheric data acquisition and transmission using a weather balloon platform. The system combines sensing, processing, power management, and wireless communication into a unified structure to achieve efficient environmental monitoring.

At the core of the architecture is the ESP32-WROOM-32, which functions as both the data processing unit and communication interface. It acquires analog and digital signals from multiple sensors, performs necessary processing, and transmits the data over Wi-Fi to a cloud-based server. The use of an ESP32 enables seamless integration of sensing and communication within a single module, reducing system complexity.

The sensing subsystem consists of three primary sensors: BMP280, DHT11, and MQ-135. These sensors are interfaced with the ESP32 through appropriate communication protocols such as I2C and digital input. The BMP280 provides pressure-based altitude estimation along with temperature measurement, while the DHT11 supplies humidity data. The MQ-135 sensor outputs analog signals corresponding to air quality parameters, which are processed by the microcontroller.

The power subsystem is designed to ensure stable operation under varying conditions. A 3.7V Li-ion battery serves as the primary power source, and its output is regulated using DC-DC boost converters (XL6009 and MT3608) to meet the voltage requirements of the ESP32 and sensors. A TP4056 charging module is incorporated for controlled battery charging and protection. Passive components such as resistors and capacitors are included to enhance signal stability and reduce noise.

The communication subsystem utilizes the built-in Wi-Fi of the ESP32 to transmit data to the cloud platform, specifically the Blynk server. This enables real-time data logging and remote visualization through a mobile application.

3.2 System Architecture

The system architecture of the proposed weather balloon is designed as a multi-layer embedded IoT system, integrating sensing, processing, power management, and communication subsystems. The architecture ensures efficient data acquisition, stable operation, and real-time transmission of atmospheric parameters.

1. **Overall Architecture Description:**

   The system follows a centralized architecture, where the ESP32-WROOM-32 acts as the core controller. All sensors are interfaced directly with the ESP32, which processes the data and transmits it to the cloud via Wi-Fi.

   The architecture can be divided into four main subsystems:

   - Sensing Subsystem
   - Processing Subsystem
   - Power Management Subsystem
   - Communication Subsystem

2. **Block Diagram Representation:**

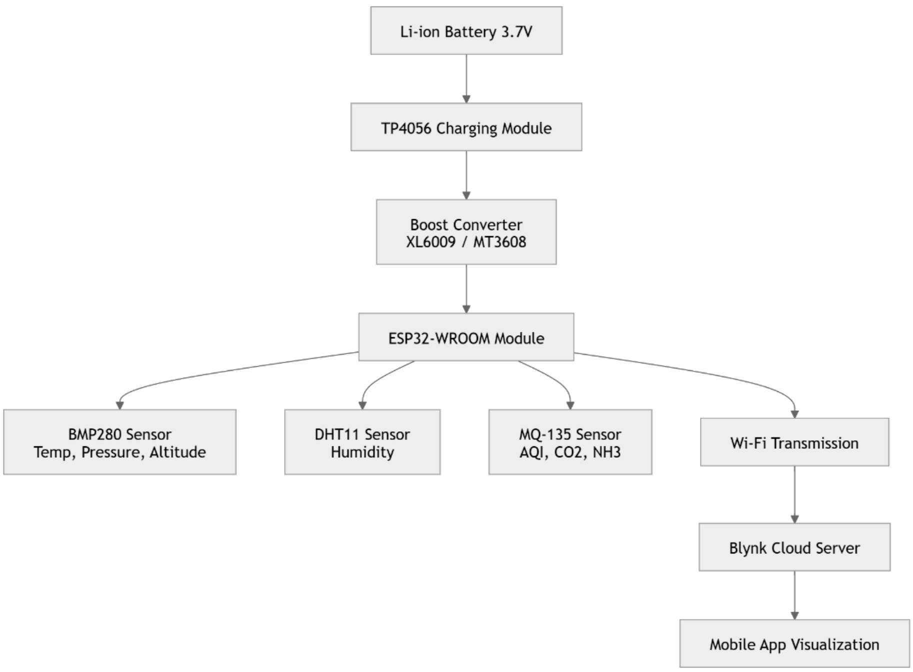

Fig 3.2 : System architecture Block Diagram

3. Subsystem Description:

a) Sensing Subsystem

The sensing unit consists of three sensors:

- BMP280: Measures temperature, pressure, and calculates altitude using pressure variation
- DHT11: Provides humidity data
- MQ-135: Detects air quality parameters such as AQI, CO_2, and NH_3.

b) Processing Subsystem

The ESP32-WROOM-32 serves as the processing unit. It:

- Reads sensor data
- Performs necessary calculations
- Formats the data for transmission
- Controls communication with the cloud

c) Power Management Subsystem

The system is powered using a 3.7V Li-ion battery. Since this voltage is not sufficient for stable operation:

- XL6009 and MT3608 boost converters are used to step up the voltage
- A TP4056 charging module ensures safe battery charging
- A 470µF capacitor stabilizes voltage fluctuations

d) Communication Subsystem

The ESP32 uses its built-in Wi-Fi module to:

- Connect to a mobile hotspot
- Send data to the Blynk cloud server
- Enable real-time monitoring through a mobile application

4. Data Flow Explanation

The system operates in a sequential manner:

1. Sensors collect environmental data
2. Data is sent to ESP32
3. ESP32 processes and formats data
4. Data is transmitted via Wi-Fi
5. Cloud server stores and processes data
6. User views data on mobile app

5. **Architectural Features**

i. Modular design for easy expansion
ii. Wireless communication eliminates need for ground station
iii. Low power consumption suitable for balloon deployment
iv. Real-time data transmission using IoT

3.3 Components used in Proposed System

3.3.1. ESP32-WROOM-32 Microcontroller

The ESP32-WROOM-32 serves as the **central processing and communication unit** of the system. It is responsible for interfacing with all sensors, acquiring environmental data, and performing necessary processing operations. The ESP32 supports both analog and digital inputs, allowing seamless integration with sensors like MQ-135 (analog) and BMP280/DHT11 (digital). Its built-in Wi-Fi capability enables direct communication with cloud platforms without requiring external modules. In this project, it connects to a mobile hotspot and transmits real-time data to the Blynk cloud server. The ESP32 also handles timing, data formatting, and synchronization between different subsystems. Its low power consumption, compact size, and high processing capability make it ideal for battery-powered applications such as weather balloon payloads. Overall, it acts as the **brain of the system**, coordinating sensing, processing, and communication tasks efficiently.

Fig-3.3.1 : esp32 WROOM 32

3.3.2. BMP280 Sensor:

The BMP280 sensor is used to measure temperature, atmospheric pressure, and altitude. It operates using the I2C communication protocol, allowing efficient data transfer to the ESP32. The sensor measures pressure variations in the atmosphere, which are then used to calculate altitude based on standard atmospheric equations. This makes it particularly useful in a weather balloon system where altitude changes continuously during ascent. Additionally, it provides temperature readings that help in analyzing environmental conditions at different heights. The BMP280 is known for its high accuracy and stability compared to basic sensors, making it suitable for atmospheric monitoring applications. In this project, it plays a crucial role in tracking vertical environmental variations and provides essential data for understanding pressure and altitude relationships. Its compact size and low power consumption further enhance its suitability for embedded and portable systems.

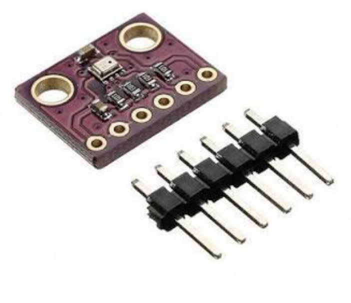

Fig-3.3.2: BMP280 sensor

3.3.3. DHT11 Sensor

The DHT11 sensor is used to measure relative humidity in the surrounding environment. It provides digital output, which simplifies interfacing with the ESP32 microcontroller. Although it is a low-cost sensor with moderate accuracy, it is sufficient for basic environmental monitoring and educational purposes. The sensor operates by detecting moisture levels in the air and converting them into electrical signals, which are then processed by the ESP32. In this project, the DHT11 complements the BMP280 by adding humidity data to the overall atmospheric dataset. This combination allows a more comprehensive analysis of environmental conditions. The sensor is easy to use, requires minimal external components, and operates at low power, making it suitable for battery-powered applications like weather balloons.

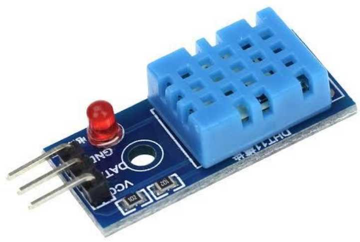

Fig- 3.3.3 : DHT11 sensor

3.3.4. MQ-135 Sensor

The MQ-135 sensor is used for air quality monitoring, detecting gases such as carbon dioxide (CO_2), ammonia (NH_3), and other pollutants. It outputs an analog signal that varies with gas concentration, which is read by the ESP32's analog input pins. The sensor requires proper calibration to provide accurate readings, as its output depends on environmental conditions and sensor aging. In this project, the MQ-135 is used to estimate Air Quality Index (AQI) and monitor pollution levels at different altitudes. This adds an important dimension to the system by enabling environmental pollution analysis along with basic atmospheric parameters. The sensor consumes relatively higher power compared to other sensors due to its internal heating element, which must be considered in power design. Despite this, it is widely used due to its affordability and ability to detect multiple gases, making it suitable for experimental and educational applications.

Fig -4: MQ-135 sensor

3.3.5. Li-ion Battery (3.7V)

The system is powered by a 3.7V Li-ion rechargeable battery, which provides a compact and lightweight energy source suitable for aerial applications. The battery ensures portability and eliminates the need for external power connections during operation. In this project, it supplies power to the ESP32, sensors, and other electronic components. However, since the nominal voltage of 3.7V is not sufficient for stable operation of all components, additional voltage regulation is required. The battery's high energy density allows longer operation time while keeping the payload weight minimal, which is critical for weather balloon deployment. Proper battery management is essential to prevent overcharging, deep discharge, and ensure safe operation. Overall, the Li-ion battery acts as the primary power source, enabling autonomous functioning of the system during flight.

Fig-3.3.5 : Li-Ion battery (3.7V)

3.3.6. Boost Converters (XL6009 & MT3608)

The boost converters, specifically XL6009 and MT3608, are used to step up the battery voltage to required levels for stable system operation. Since the Li-ion battery provides 3.7V, which may not be sufficient for consistent performance of the ESP32 and sensors, these converters increase the voltage to a regulated output. The use of two boost converters helps distribute power efficiently across different components if needed. These converters ensure that voltage fluctuations do not affect system performance, especially during high current demand. They are adjustable, allowing precise control of output voltage, and are known for their efficiency and reliability. In this project, they play a critical role in maintaining a stable power supply, which is essential for accurate sensor readings and uninterrupted communication. Without proper voltage regulation, the system could experience instability or malfunction.

Fig-3.3.6 MT3608 boost converter

Fig-3.3.6: XL6009 boost converter

3.3.7. TP4056 Charging Module

The TP4056 is used for safe charging and protection of the Li-ion battery. It provides features such as overcharge protection, over-discharge protection, and short-circuit protection. The module allows the battery to be charged via a USB power source, making it convenient and user-friendly. In this project, it ensures that the battery operates within safe limits, thereby increasing its lifespan and reliability. The TP4056 regulates the charging current and voltage, preventing damage to the battery. It is an essential component in portable systems where rechargeable batteries are used. By integrating this module, the system achieves efficient energy management and enhances safety during repeated charging cycles.

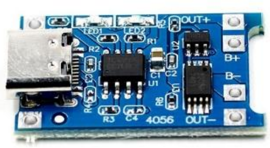

Fig-3.3.7: TP4056 Charging module

3.3.8. Passive Components (Resistors & Capacitor)

Passive components such as 10kΩ resistors and a 470µF capacitor are used to ensure proper circuit operation and stability. The resistors are used for purposes such as pull-up/pull-down configurations, signal conditioning, and limiting current where necessary. The capacitor plays a crucial role in voltage stabilization, filtering out noise and reducing fluctuations in the power supply. This is particularly important in a system with multiple sensors and wireless communication, where stable voltage is required for accurate readings and reliable operation. These components, although simple, are essential for maintaining the integrity of the circuit and preventing unwanted disturbances.

Fig-3.3.8: 10kilo ohms Resistor

Fig-3.3.8: 470micro Farads

3.4 Working Principle

The working principle of the proposed weather balloon system is based on the integration of sensing, processing, and wireless communication to enable real-time atmospheric monitoring. The system operates in a sequential manner, starting from power supply initialization to data visualization on a mobile device.

Initially, the system is powered by a 3.7V Li-ion battery, and the required operating voltage is achieved using boost converters (XL6009 and MT3608). The regulated power is supplied to the ESP32-WROOM-32 and all connected sensors. Once powered, the ESP32 initializes the sensors and establishes a Wi-Fi connection using a mobile hotspot.

The sensing stage involves collecting environmental data through multiple sensors. The BMP280 sensor measures temperature, atmospheric pressure, and calculates altitude based on pressure variations. The DHT11 sensor measures relative humidity, while the MQ-135 sensor detects air quality parameters such as AQI, CO_2, and NH_3 by generating analog signals corresponding to gas concentrations.

The ESP32 continuously reads data from these sensors at regular intervals. It processes and formats the data into suitable values for transmission. The processed data is then sent via Wi-Fi to the cloud platform, specifically the Blynk server.

On the cloud side, the data is stored and updated in real time. The user can access this data through the Blynk mobile application, where it is displayed in graphical or numerical form. This allows continuous remote monitoring of atmospheric conditions without physically accessing the device.

Thus, the system follows a complete cycle of data acquisition → processing → transmission → visualization, enabling efficient and real-time environmental monitoring during the balloon’s operation.

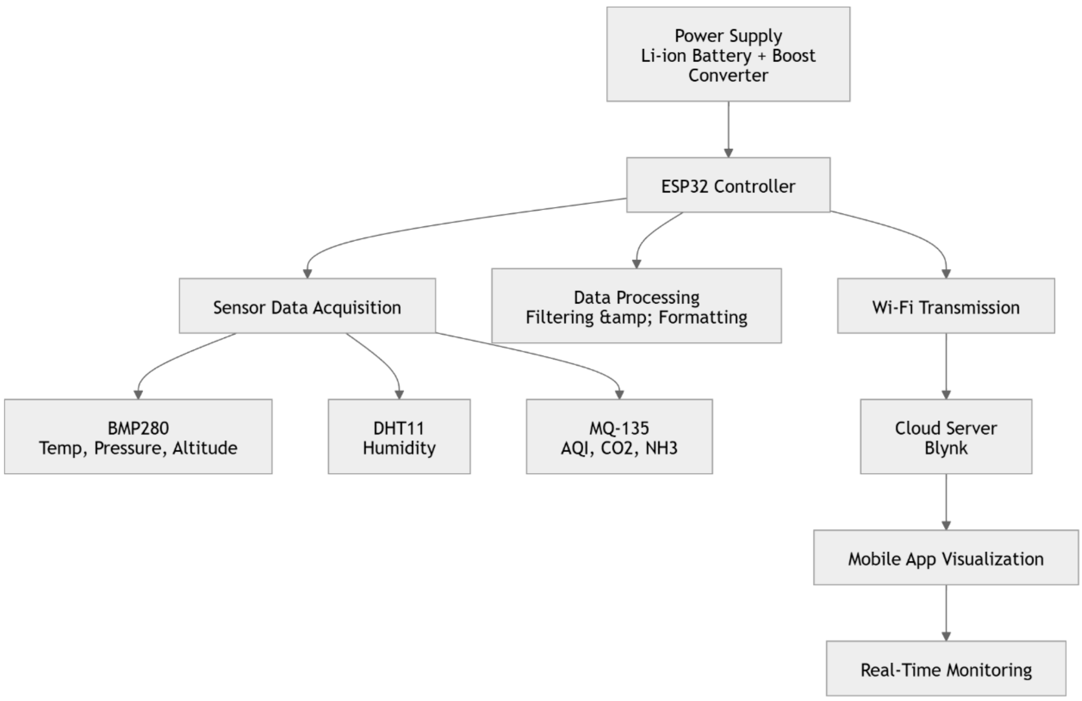

Fig-3.4 : Working Block Diagram

CHAPTER-4
RESULTS AND ANALYSIS

4.1 Technologies and Tools Used

The development of the proposed weather balloon system involves a combination of embedded systems, IoT platforms, power electronics, and fabrication tools. These technologies collectively enable efficient data acquisition, processing, transmission, and physical integration of the system.

4.1.1. Embedded System Technology

The project is built around the ESP32-WROOM-32, which serves as the core embedded platform. It is programmed using the Arduino IDE, allowing easy development, debugging, and uploading of code. The ESP32 handles sensor interfacing, data processing, and Wi-Fi communication. Embedded C/C++ is used for writing the program logic, including sensor data acquisition, calibration, and transmission to the cloud.

4.1.2. Sensor Technology

Multiple sensors are used for environmental monitoring:

- BMP280 for temperature, pressure, and altitude
- DHT11 for humidity
- MQ-135 for air quality (AQI, CO_2, NH_3).

These sensors utilize communication protocols such as I2C (BMP280) and digital/analog interfacing (DHT11 and MQ-135). Proper sensor integration enables accurate and continuous data collection.

4.1.3. IoT and Cloud Technology

The system uses the Blynk platform for real-time data monitoring. The ESP32 connects to a mobile hotspot via Wi-Fi and transmits sensor data to the Blynk cloud server. The mobile application provides a user-friendly interface for visualizing data in real time through graphs and indicators.

4.1.4. Power Electronics

Efficient power management is achieved using:

- 3.7V Li-ion battery as the primary power source
- Boost converters (XL6009 and MT3608) for voltage regulation
- TP4056 charging module for safe battery charging

4.1.5. 3D Printing Technology

A 3D-printed enclosure (case) is used to house and protect the electronic components. The design is created using CAD software and fabricated using a 3D printer.

This enclosure provides:

- Structural support
- Protection from environmental exposure
- Compact arrangement of components

4.1.6. Hardware Assembly Tools

Various tools are used for circuit assembly and testing:

- Breadboard and jumper wires for prototyping
- Soldering tools for permanent connections
- Multimeter for testing voltage and continuity

4.1.7. Development and Testing Tools

- Arduino IDE for coding and debugging
- Serial Monitor for real-time data checking
- Blynk dashboard for cloud-based visualization.

4.2 Hardware Specifications

The hardware specifications of the proposed system define the electrical, functional, and operational characteristics of each component used in the weather balloon payload. These specifications ensure proper selection, compatibility, and reliable performance of the system during operation.

4.2.1. Microcontroller Specifications

Parameter	**Specification**
Module	ESP32 WROOM-32
Operating Voltage	3.0V – 3.6V
Input Voltage (Vin)	5V
Clock Frequency	Up to 240 MHz
Wi-Fi	802.11 b/g/n
GPIO Pins	34 Pins
ADC Resolution	12-bit
Communication Protocols	I2C, SPI, UART

Table – 4.2.1.1 Microcontroller Specifications

2. Sensor Specifications

a) BMP280 Sensor

Parameter	**Specification**
Operating Voltage	1.8V – 3.6V

Parameter	Specification
Pressure Range	300 hPa – 1100 hPa
Temperature Range	-40°C to +85°C
Interface	I2C / SPI
Accuracy	High precision

Table – 4.2.2.1 BMP280 Sensor

b) DHT11 Sensor

Parameter	Specification
Operating Voltage	3V – 5V
Humidity Range	20% – 90% RH
Temperature Range	0°C to 50°C
Accuracy	±5% RH
Output Type	Digital

Table – 4.2.2.2 DHT11 Sensor

3. Power Supply Specifications

a) Li-ion Battery

Parameter	**Specification**
Nominal Voltage	3.7V
Type	Rechargeable Lithium-ion
Capacity	Depends on model (e.g., 2000–3000 mAh)

Table – 4.2.3.1 Li-ion Battery

b) Boost Converters

Parameter	**XL6009 / MT3608**
Input Voltage	2V – 24V
Output Voltage	Adjustable (up to ~28V)
Efficiency	Up to 90%
Function	Step-up voltage conversion

Table – 4.2.3.2 Boost Converters

c) TP4056 Charging Module

Parameter	**Specification**
Charging Voltage	4.2V

Parameter	Specification
Input Voltage	5V (USB)
Protection Features	Overcharge, over-discharge, short circuit

Table – 4.2.3.3 TP4056

5. Physical and Structural Components

Component	Specification
Enclosure	3D Printed Case
Material	PLA / ABS (depending on printing)
Function	Protection and structural support
Weight Consideration	Lightweight for balloon payload

Table – 4.2.5 Physical and Structural Components

6. Connectivity Components

Component	Specification
Jumper Wires	Male-to-Male / Male-to-Female
Function	Circuit connections

Table – 4.2.6 Connectivity Components

PIN TO PIN CONNECTIONS TABLE:

Component	Pin	Connect To ESP32
BME280	VCC	3.3V
	GND	GND
	SDA	GPIO21
	SCL	GPIO22
MQ135	VCC	VIN / 5V (From the Boost Converter MT3608)
	GND	GND
	A0	GPIO34
DHT11	VCC	3.3V
	GND	GND
	DATA	GPIO4
Li-Po Battery	+	VIN
	–	GND
TP4056	+ Out	Battery (+)
	- Out	Battery (-)

Table : 4.2.7 PIN TO PIN CONNECTIONS

Note:

- **ALL GND’s ARE COMMON**

4.3 Software Specifications

The software specifications define the programming environment, communication protocols, libraries, and functional behaviour of the proposed weather balloon system. The software is responsible for sensor data acquisition, processing, wireless communication, and cloud integration.

1. Development Environment

Parameter	Specification
IDE	Arduino IDE
Programming Language	Embedded C / C++
Platform Support	Windows / Linux
Board Configuration	ESP32 Board Package

Table – 4.3.1 Development Environment

2. Microcontroller Programming

The ESP32 is programmed to perform the following functions:

- Initialization of sensors and peripherals
- Establishing Wi-Fi connection using hotspot credentials
- Reading sensor data at regular intervals
- Processing and formatting data
- Sending data to the cloud server

3. Libraries Used

Library Name	Purpose
WiFi.h	Wi-Fi connectivity
BlynkSimpleEsp32.h	Cloud communication with Blynk

Library Name	**Purpose**
Adafruit BMP280	Interface with BMP280 sensor
DHT.h	Interface with DHT11 sensor
MQ135 (or analog read)	Air quality measurement

Table – 4.3.2 Libraries Used

4. Communication Protocols

I2C	BMP280 sensor communication
Digital I/O	DHT11 data communication
Analog Input	MQ-135 data acquisition
Wi-Fi	Data transmission to cloud

Table – 4.3.3 Communication Protocols

5. IoT Platform Integration

The system uses the Blynk platform for:

- Real-time data visualization
- Remote monitoring via mobile app
- Cloud data storage

6. Code Used in project

```
#define BLYNK_PRINT Serial
#define BLYNK_TEMPLATE_ID "TMPL3HUO_Rf2A"
#define BLYNK_TEMPLATE_NAME "WEATHER MONITORING SYSTEM"
#define BLYNK_AUTH_TOKEN "dJNV30B6BTQIC3cIDOFCbkf24CSn6olp"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_BMP280.h>
#include <DHT.h>

// -------- WiFi --------
```

```
char ssid[] = "WIFI-NAME";
char pass[] = "WIFI-PASSWORD";

// -------- Sensors --------
Adafruit_BMP280 bmp;

#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

#define MQ135_PIN 34

// MQ calibration (adjust later)
float R0 = 10.0;

BlynkTimer timer;

// -------- Moving Average --------
#define FILTER_SIZE 10
float mqBuffer[FILTER_SIZE];
int mqIndex = 0;

// -------- ADC Setup --------
void setupADC() {
  analogReadResolution(12);
  analogSetAttenuation(ADC_11db);
}

// -------- MQ135 Resistance --------
float getResistance(int adc) {
  float voltage = adc * (3.3 / 4095.0);
  if (voltage <= 0.01) return 0;
  return ((3.3 - voltage) / voltage) * 10.0;
}

// -------- Filtered ADC --------
int getFilteredADC() {
  int raw = analogRead(MQ135_PIN);

  mqBuffer[mqIndex] = raw;
  mqIndex = (mqIndex + 1) % FILTER_SIZE;

  float sum = 0;
  for (int i = 0; i < FILTER_SIZE; i++) {
    sum += mqBuffer[i];
  }

  return sum / FILTER_SIZE;
```

```
}

// -------- MAIN FUNCTION --------
void sendData() {

  Serial.println("------ SENSOR DATA ------");

  // ===== BMP280 =====
  float temp_bmp = bmp.readTemperature();
  float pressure = bmp.readPressure() / 100.0F;
  float altitude = bmp.readAltitude(1013.25);

  if (!isnan(temp_bmp) && !isnan(pressure)) {
    Blynk.virtualWrite(V0, temp_bmp);
    Blynk.virtualWrite(V1, pressure);
    Blynk.virtualWrite(V2, altitude);

    Serial.printf("BMP Temp: %.2f | Pressure: %.2f | Alt: %.2f\n",
                  temp_bmp, pressure, altitude);
  } else {
    Serial.println("BMP280 ERROR");
  }

  // ===== DHT11 =====
  float humidity = dht.readHumidity();
  float temp_dht = dht.readTemperature();

  if (!isnan(humidity) && !isnan(temp_dht)) {
    Blynk.virtualWrite(V3, temp_dht);
    Blynk.virtualWrite(V4, humidity);

    Serial.printf("DHT Temp: %.2f | Humidity: %.2f\n",
                  temp_dht, humidity);
  } else {
    Serial.println("DHT11 ERROR");
  }

  // ===== MQ-135 (REAL) =====
  int adc = getFilteredADC();
  float rs = getResistance(adc);

  if (rs > 0) {
    float ratio = rs / R0;

    // CO2 approximation
    float co2 = 116.6020682 * pow(ratio, -2.769034857);

```

```
    // Clamp realistic range
    if (co2 < 350) co2 = 350;
    if (co2 > 5000) co2 = 5000;

    // AQI mapping
    float aqi = map(co2, 400, 2000, 50, 300);
    if (aqi < 0) aqi = 0;
    if (aqi > 500) aqi = 500;

    Blynk.virtualWrite(V5, aqi);
    Blynk.virtualWrite(V6, co2);

    Serial.printf("ADC: %d | RS: %.2f | CO2: %.2f ppm | AQI: %.2f\n",
                  adc, rs, co2, aqi);
  } else {
    Serial.println("MQ135 ERROR");
  }

  Serial.println("--------------------------\n");
}

// -------- SETUP --------
void setup() {
  Serial.begin(115200);
  delay(1000);

  setupADC();
  Wire.begin(21, 22);
  dht.begin();

  // BMP init
  if (!bmp.begin(0x76)) {
    if (!bmp.begin(0x77)) {
      Serial.println("BMP280 NOT FOUND!");
      while (1);
    }
  }

  Serial.println("Sensors initialized");

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

  timer.setInterval(2000L, sendData);
}

// -------- LOOP --------
void loop() {
```

```
        Blynk.run();
        timer.run();
    }
```

4.4 Experimental Results

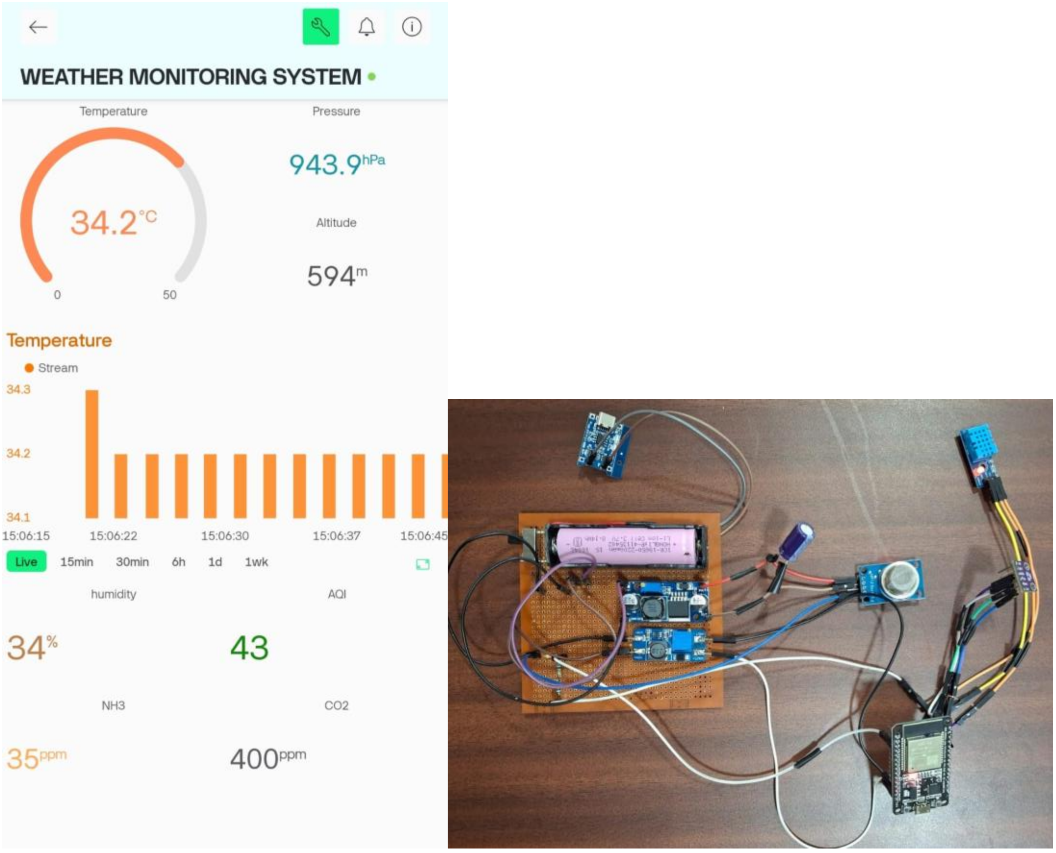

Fig- 4.4 Experimental Results

The experimental results of the proposed weather balloon system were obtained by testing the setup under normal environmental conditions before deployment. The system was powered using a Li-ion battery, and all sensors were interfaced with the ESP32-WROOM-32. Data was successfully transmitted to the cloud using the Blynk application.

During testing, the sensors provided continuous real-time readings. The BMP280 sensor recorded temperature, pressure, and altitude values, showing stable variations corresponding to environmental changes. The DHT11 sensor provided humidity readings within its expected accuracy range. The MQ-135 sensor detected air quality variations, although its readings required calibration for precise interpretation.

Sample Observations

Parameter	Observed value
Temperature	32.4 degree C
Humidity	34 %
Pressure	943.9 hPa
Altitude	594 m
NH3	35 ppm
CO2	400 ppm
Air Quality (AQI)	43

Table- 4.4 Sample Observations

4.5 Performance Evaluation

The performance of the proposed IoT-based weather balloon system is evaluated based on parameters such as accuracy, reliability, response time, power efficiency, and communication performance. The system demonstrates satisfactory operation for educational and experimental applications.

1. Sensor Performance

The sensors used in the system provide reasonably accurate readings under normal conditions:

- BMP280 shows stable and accurate measurements for temperature, pressure, and altitude. It performs well in detecting pressure variations, enabling effective altitude estimation.
- DHT11, although less precise, provides acceptable humidity readings for basic monitoring.
- MQ-135 offers qualitative air quality data; however, its accuracy depends on proper calibration and environmental conditions.

The sensor performance is suitable for approximate atmospheric analysis, though not for high-precision scientific measurements.

2. Communication Performance

The ESP32-WROOM-32 ensures reliable data transmission using Wi-Fi. Data is successfully sent to the Blynk cloud server with minimal delay under stable network conditions.

- **Latency:** Low, typically a few seconds

- **Reliability:** High within Wi-Fi coverage range
- **Limitation:** Performance degrades when signal strength is weak or unavailable.

3. Power Efficiency

The system operates on a 3.7V Li-ion battery, with boost converters ensuring stable voltage supply. The ESP32 operates efficiently in normal mode, but:

- Continuous Wi-Fi usage increases power consumption
- MQ-135 sensor consumes additional power due to its heating element

4. System Stability

The inclusion of proper power regulation components (boost converters, capacitor) ensures:

- Stable voltage supply
- Reduced noise and fluctuations
- Consistent sensor readings

The system shows stable operation without frequent resets or failures.

5. Real-Time Monitoring Performance

The integration with the Blynk allows:

- Continuous real-time data updates
- Easy visualization through graphs and indicators
- Remote accessibility from mobile devices.

6. Overall Evaluation

Parameter	**Performance Level**
Accuracy	Moderate
Reliability	Good
Power Efficiency	Moderate
Communication	Good (Wi-Fi dependent)
Scalability	High

CHAPTER-5
CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The proposed IoT-based weather balloon system successfully demonstrates a cost-effective and compact solution for real-time atmospheric monitoring. By integrating the ESP32-WROOM-32 with multiple sensors such as BMP280, DHT11, and MQ-135, the system is capable of measuring key environmental parameters including temperature, humidity, pressure, altitude, and air quality.

The use of Wi-Fi communication and cloud integration through the Blynk enables real-time data transmission and remote visualization, eliminating the need for complex ground station setups. The incorporation of a Li-ion battery, boost converters, and a charging module ensures efficient power management, making the system portable and suitable for balloon deployment.

Although the system has certain limitations, such as dependency on Wi-Fi connectivity and moderate sensor accuracy, it effectively meets the project objectives. The project also provides valuable hands-on experience in embedded systems, IoT, and sensor integration.

Overall, the system proves that reliable atmospheric monitoring can be achieved using affordable and easily available components, making it highly suitable for educational and experimental applications.

5.2 Future Scope

The proposed system offers several opportunities for enhancement and expansion to improve its performance and applicability:

- **Integration of GPS Module:**
  Adding a GPS module will enable real-time tracking of the balloon's location and altitude, improving data correlation.
- **Use of Long-Range Communication (LoRa):**

Replacing Wi-Fi with LoRa technology can significantly extend communication range, making the system suitable for high-altitude applications.

- **Improved Sensor Accuracy:**
  Replacing DHT11 and MQ-135 with higher-precision sensors can enhance measurement reliability.
- **Data Logging and Storage:**
  Incorporating onboard storage (e.g., SD card module) will allow data backup even when connectivity is lost.
- **Enhanced Power Management:**
  Optimizing power consumption using sleep modes and efficient components can increase operational time.
- **Advanced Data Analysis:**
  Integration with data analytics platforms can enable trend analysis and predictive modelling.
- **Improved Enclosure Design:**
  Designing a more robust and insulated 3D-printed case can protect the system from extreme environmental conditions.
- **Multi-Node Deployment:**
  Expanding the system to multiple balloon units can enable large-scale atmospheric data collection.

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